Fundusze Europejskie
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„ZPR PWr – Zintegrowany Program Rozwoju Politechniki Wrocławskiej"

Virtual Reality,
VR support in Unity

Marek Kopel

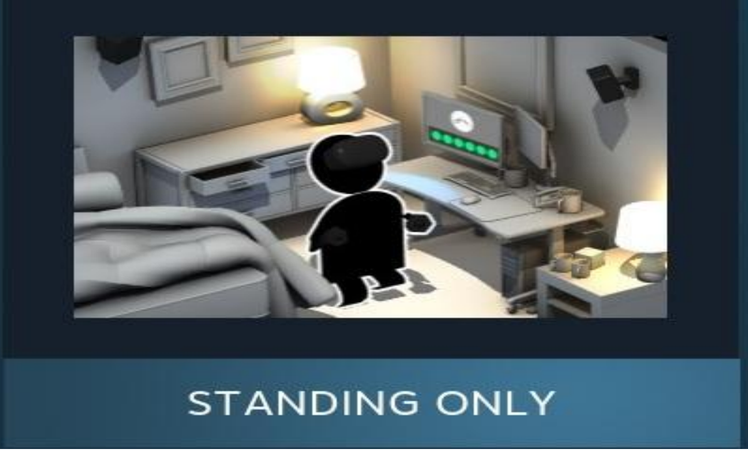
wideo z wykładu:

https://pwr-edu.zoom.us/rec/share/kxClOTxSojwun0_2veLnWDU3qCrpWIwhW1O8lHm1t5LI3_nVhurjFsYZArWyw8Be.rb0_snWVaLgzTCiB

Lecture

1. **Silniki gier. Wprowadzenie do Unity. Pierwsza gra 2D.**
2. **Mechanika 2D, animacje.**
3. **Oświetlenie, tekstury, materiały. Pierwsza gra 3D.**
4. Narzędzia wspierające, np. Blender.
5. **Zapis/odczyt danych, komunikacja sieciowa.**
6. **Modelowanie terenu. Generowanie terenu.**
7. **Sztuczna inteligencja w grach.**
8. Virtual Reality, wsparcie VR w Unity.
9. **Optymalizacje w Unity.**
10. Historia i klasyfikacja gier.
11. Prototypowanie gier. GDD.
12. Projektowanie poziomów gier.
13. Projektowanie gier dla różnych platform.
14. Testowanie gier. Test.


ROOM-SCALE


STANDING ONLY


VR Gaming Guide
Sitting Down
Standing
Room Scale

Ile to kosztuje?

cardboard

stand-alone HMD

HMD + desktop

	Price	Headsets available
Cheap	$15 - $50	Smaller brands, usually running off smartphones
Mid-range	$300 - $600	Oculus Quest 2, HP Reverb G2
Premium	$800 - $1,000	HTC Vive Pro 2, Valve Index

Meta Oculus Quest 2 - **1600 PLN**

HTC Vive Pro 2 goggle - **3800 PLN**, cały zestaw (bez PC) - **6700 PLN**

Sony PlayStation VR2 - **3000 PLN**, z konsolą - **7000 PLN**

How much is a VR headset? | GamesRadar+

The best VR headset in 2023: all the latest platforms compared | GamesRadar+

XR

https://docs.unity3d.com/Manual/XR.html

XR is an *umbrella term* that includes the following types of applications:

Virtual Reality (VR): The application simulates a completely different environment around the user.

Mixed Reality (MR): The application combines its own environment with the user's real-world environment and allows them to interact with each other.

Augmented Reality (AR): The application layers content over a digital view of the real world.

Unity supported platforms
https://docs.unity3d.com/Manual/XR.html

Unity supports these platforms for:

ARKit
ARCore
Microsoft HoloLens
Windows Mixed Reality
Magic Leap
Oculus
OpenXR
PlayStation VR

Unity **does not** support XR on **WebGL**

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Unity supported platforms
https://docs.unity3d.com/Manual/VROverview.html

Oculus

If you're developing for **Oculus Quest**, you need to install and load the **Oculus XR Plug-in for the Android build target**.
If you're developing for **Oculus Rift and Rift S**, you need to install and load the **Oculus XR Plug-in for the Windows build target**.

Windows Mixed Reality

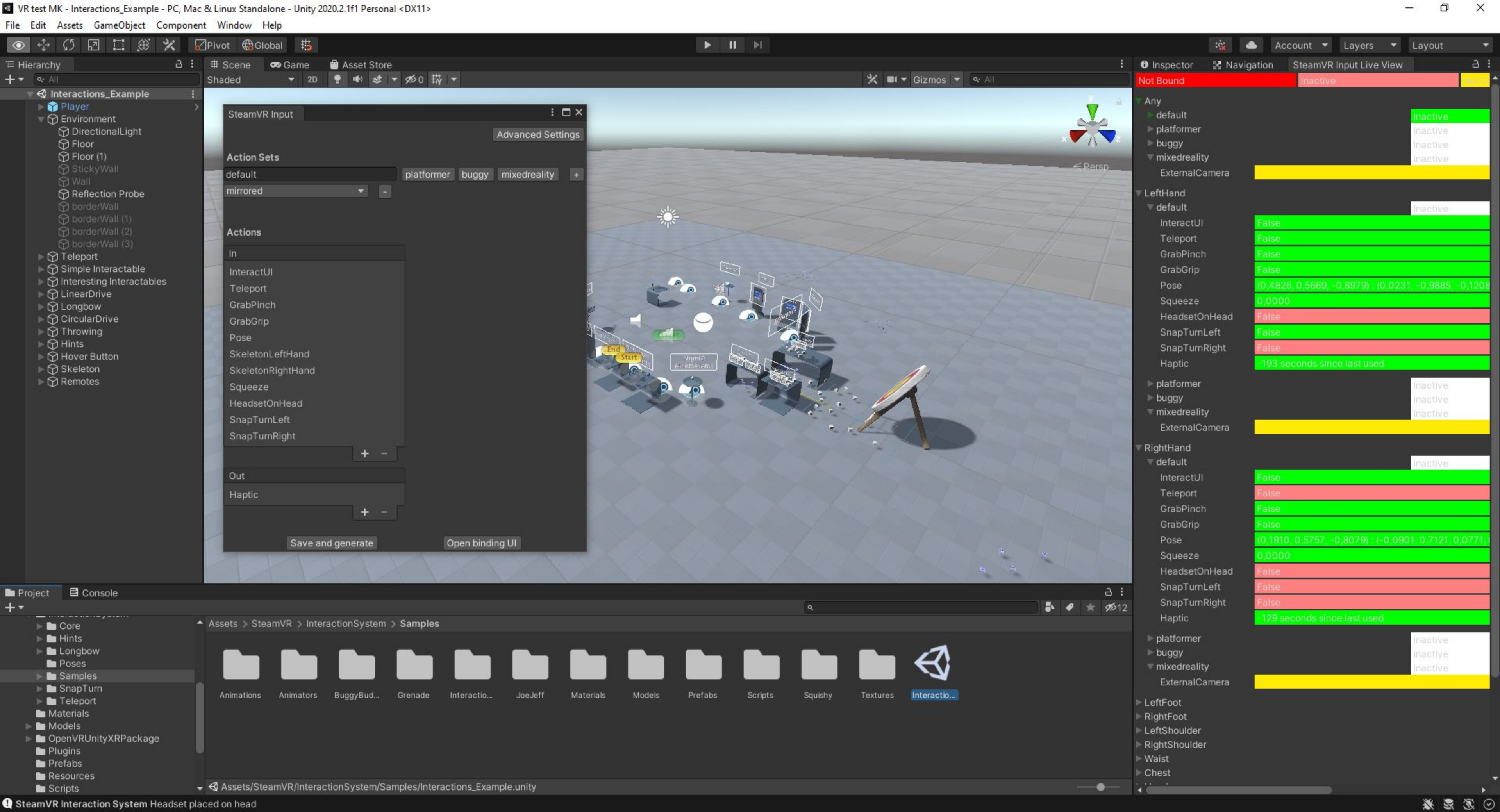
If you're developing for a **Windows MR VR headset**, you need to install and load the **Windows XR Plug-in for the Universal Windows Platform**, and use Microsoft's Mixed Reality Toolkit (MRTK 2.3) to add mixed reality input and interactions.

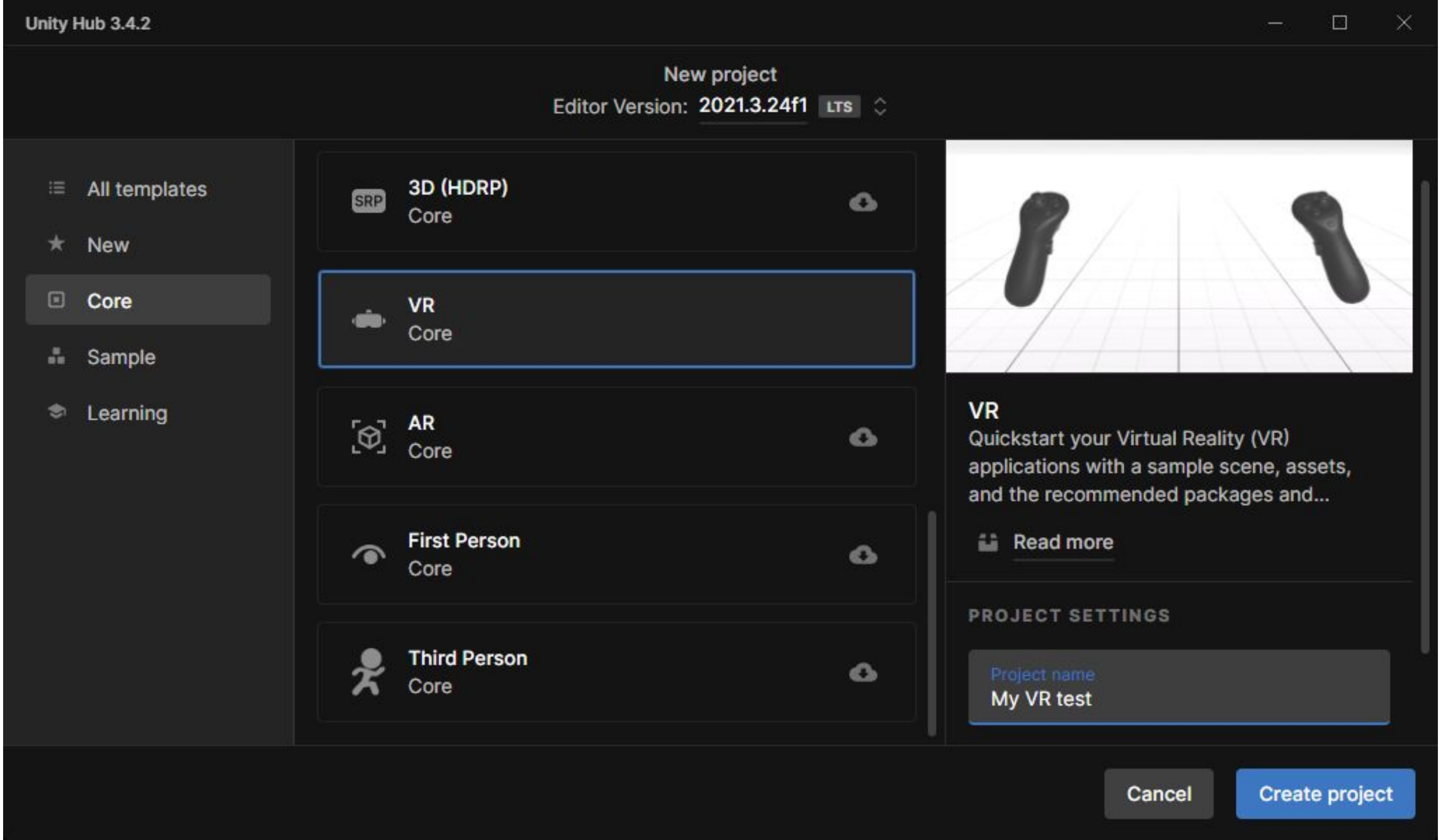
Third-party supported platforms

Google VR ⟶ Google's VR SDK Quickstart guide for the latest on Google Daydream and Cardboard

OpenVR ⟶ Valve has released their OpenVR Unity XR Plug-in → **SteamVR (HTC Vive, Valve Index)**

https://assetstore.unity.com/packages/tools/integration/steamvr-plugin-32647




Unity Hub 3.4.2
New project
Editor Version: 2021.3.24f1 LTS
All templates
New
Core
Sample
Learning
SRP
3D (HDRP)
Core
VR
Core
AR
Core
First Person
Core
Third Person
Core
VR
Quickstart your Virtual Reality (VR) applications with a sample scene, assets, and the recommended packages and...
Read more
PROJECT SETTINGS
Project name
My VR test
Cancel
Create project

VR vs non-VR

jedyny aspekt, gdzie wygrał non-VR.

2. How **enjoyable and engaging** were the sensations of **moving, turning, crouching, hiding**?

Dlaczego?

VR locomotion: treadmill vs skating

12 VR Locomotion Solutions/Concepts You Probably Haven't Heard Of

EKTO ONE Product Reveal (covered above)

A completely new way to walk in VR: CyberShoes (8:15)

- problems with shoes and full body tracking

https://www.facebook.com/reel/314097304394223

Oculus in Unity

https://assetstore.unity.com/packages/tools/integration/oculus-integration-82022

→ https://developer.oculus.com/documentation/unity/unity-package-manager/

Wroclaw University of Science and Technology

study case:

"How to handle head collisions in VR"

tylko **ZAJAWKA**
- całość na
Gry Komputerowe
II stopień,
spec. PSI

findings from an experiment
for a Master Thesis

by student Bartłomiej Stanasiuk
and me as supervisor

[#6 Kopel_Marek, IEAAIE2020]()

study case:

“Which gameplay aspects impact the immersion in virtual reality games?”

tylko **ZAJAWKA**
- całość na
Gry Komputerowe
II stopień,
spec. PSI

findings from an experiment
for a Master Thesis

by student Marta Rutkowska
and me as supervisor

#163 Kopel_Marek, ICCCI2021